The Artificial Neuron KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: adds compares Fighter multiplies Wizard

You can use a word more than once.

1. Fighter or Wizard? These creatures are made up.

Krellis: Strength 4, Magic 2. It's a Fighter. Lullom: Strength 2, Magic 4. It's a Wizard.

The rule I used: Answers vary. (More Strength than Magic: Fighter.)

2. Multiply, add, compare.

An artificial neuron multiplies each input by its weight, adds them up, and compares the total with the threshold.

Strength 5, Magic 2. Weights: Strength 1, Magic -1. Threshold: 0. More than the threshold: Fighter.

Total = $(5 \times 1) + (2 \times -1) = 5 - 2 = 3$ Fighter or Wizard? Fighter

Strength 2, Magic 3. Weights: Strength 2, Magic -1. Threshold: 2.

Total = $(2 \times 2) + (3 \times -1) = 4 - 3 = 1$ Fighter or Wizard? Wizard

3. Find the weights. Write every try. Change one number at a time.

Try	Strength weight	Magic weight	Threshold	Cards right (of 6)
Start	1	1	5	2
2				
3				
4				
5				

Our weights that got 6 of 6: Strength 1 Magic -1 Threshold 0 (one answer)

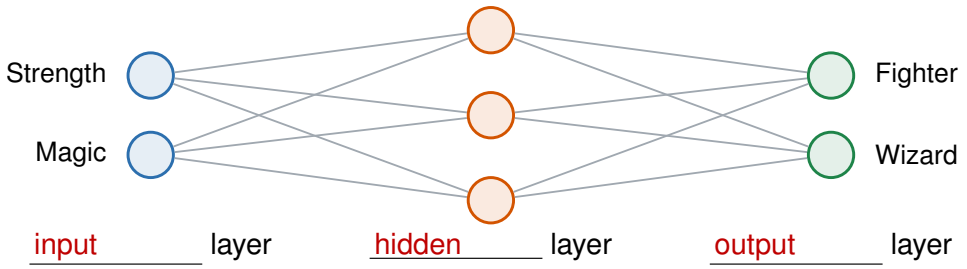
Card	Strength	Magic	Total	Neuron says	Really
1 Krellis	4	2	2	Fighter	Fighter
2 Lullom	2	4	-2	Wizard	Wizard
3 Rezivil	5	2	3	Fighter	Fighter
4 Pona	1	5	-4	Wizard	Wizard
5 Sandam	3	2	1	Fighter	Fighter
6 Skabil	3	4	-1	Wizard	Wizard

Word bank: hidden input output threshold weight
You can use a word more than once.

4. Real neuron, artificial neuron. Match each part.

Real neuron	Artificial neuron
Dendrites take signals in	input
Strong or weak synapses	weight
Fires only if the total is big enough	threshold

5. From one neuron to a network. Name each layer.



Each line is a weight. How many lines? $2 \times 3 + 3 \times 2 =$ 12

Why is the middle layer called hidden? You don't see what those neurons decide.

6. Talk about it.

We fixed the weights by Answers vary. (Testing the cards and changing one number at a time.)

I think an AI with billions of weights gets them by Answers vary. (Next lesson: it learns them.)

Go Further (Day 2). Test your neuron on all 140 Fighters and Wizards.

Our class's neuron: 91 % right. My best: 93 % with weights 4 −4 1

Can any line get all 140 right? no Why? Some Fighters have high Magic. Some Wizards are strong.